

SMART INDIA HACKATHON 2026



TITLE PAGE

- **Problem Statement ID – 26189**
- **Problem Statement Title - AI-Powered Criminal Network Analysis System**
- **Theme - Blockchain & Cybersecurity**
- **PS Category - Software**
- **Team ID - SIH2026-ST-842**
- **Team Name (Registered on portal) – Stryker**



IDEA / SOLUTION :

SENTINEL: AI-Powered Intelligence Platform for Real-Time Criminal Network Analysis & Crime Deterrence

- **Unified Knowledge Graph** : Fuses police FIRs, phone call records (CDRs), bank accounts, and video feeds into one real-time network.
- **Kingpin & Boss Discovery** : Graph centrality algorithms automatically uncover covert syndicate leaders hiding behind operatives.
- **Hawala & AML Detection** : Detects 4-node circular money loops (<72h) and smurfing transfers below ₹50,000 threshold.
- **Burner Phone Triangulation** : Tracks shared mobile handsets (IMEI reuse) and co-located nocturnal meetings (<500m).
- **CEO Prime Agent Swarm** : Autonomous multi-agent coordination commanding DataIntel, NetworkAnalyst, ThreatScoring, and Report agents.
- **Statutory Evidence Vault** : Cryptographic SHA-256 blockchain ledger ensuring 100% Section 63 BSA / 65B IEA court admissibility.

PROBLEM RESOLUTION :

❖ Breaks Data Silos :

Police usually examine FIRs, phone logs, and bank statements in isolation. SENTINEL automatically merges fragmented clues into an interactive 360° map in seconds.

❖ Exposes Hidden Syndicate Bosses :

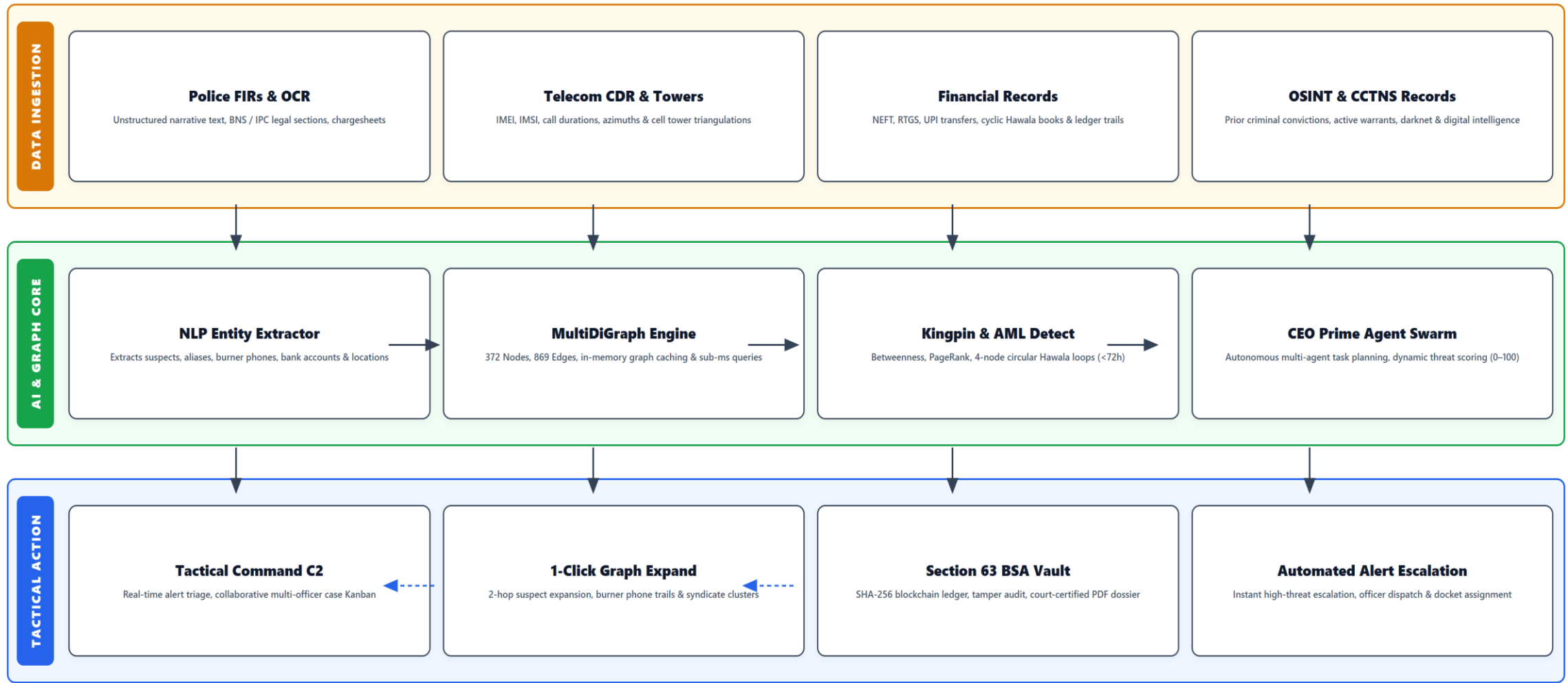
Crime kingpins rarely use their own phones. Advanced Betweenness Centrality and PageRank expose the mastermind brokers coordinating criminal activities.

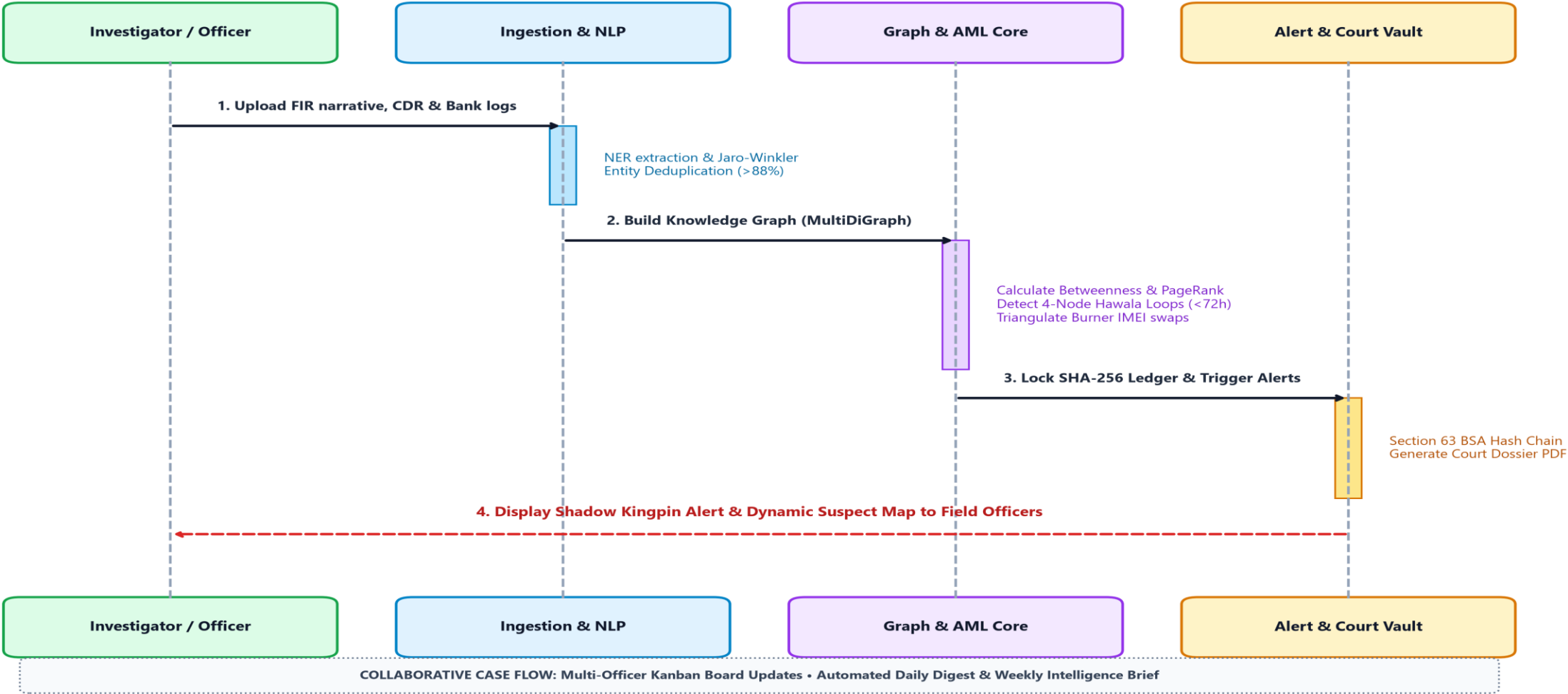
❖ Autonomous Multi-Agent Investigation :

Replaces manual investigative bottlenecks with autonomous AI agent coordination—CEO Prime Agent correlates multi-source signals, computes 0-100 threat scores, and flags high-priority syndicate targets.

❖ Guarantees Court Convictions :

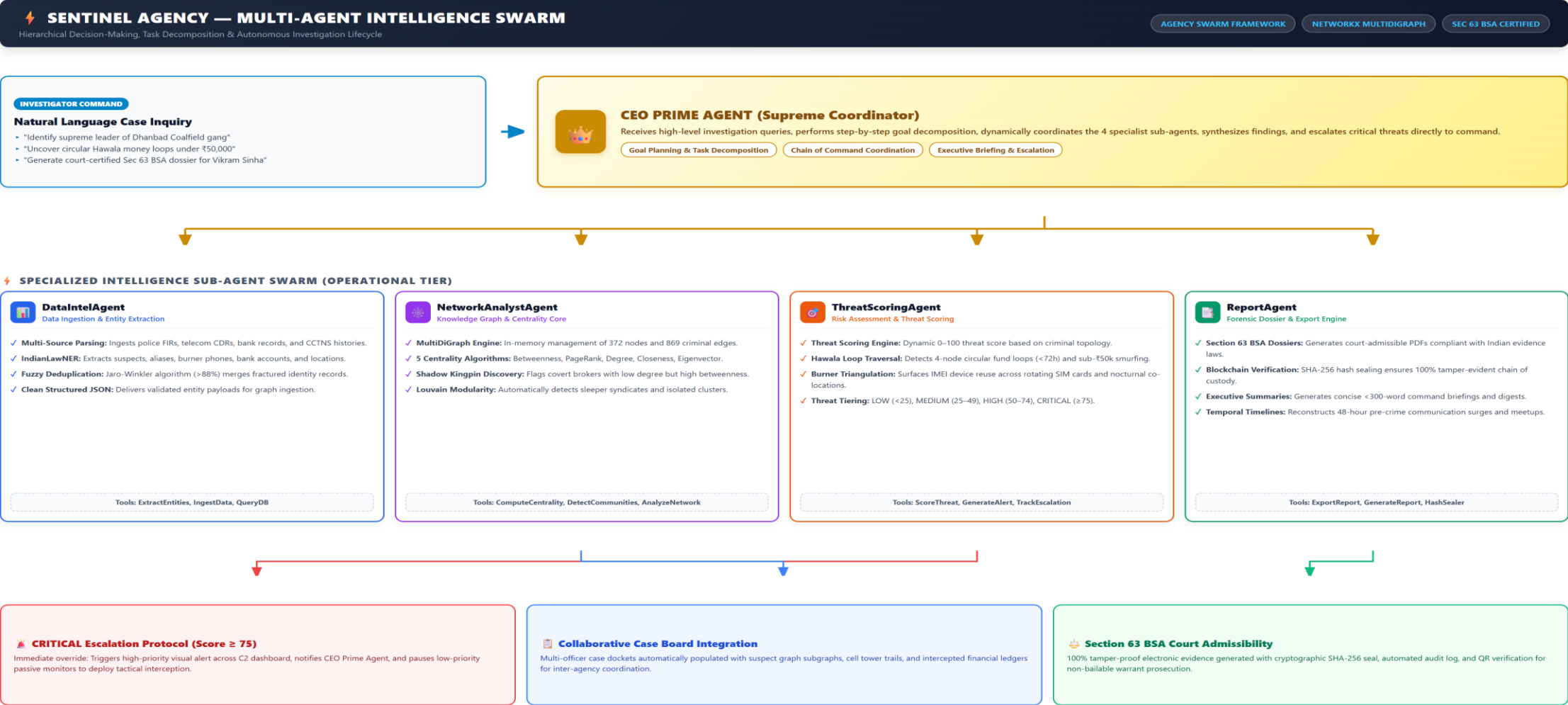
Digital evidence often fails in court due to broken chain of custody. SENTINEL creates Section 63 BSA certified PDF dossiers with SHA-256 blockchain verification.







TECHNICAL APPROACH



Frontend Dashboard :

- **React 19, TypeScript, Vite** – High-performance responsive tactical UI built for police control room command.
- **Tailwind CSS** – Low-light dark tactical theme reducing officer eye fatigue.
- **Cytoscape.js & Leaflet.js** – Interactive 2D/3D suspect graph networks and geo-spatial cell tower tracking.

Backend + Intelligence Core :

- **FastAPI (Python 3.12)** – Ultra-fast asynchronous REST APIs with sub-millisecond query latency.
- **NetworkX MultiDiGraph** – In-memory graph engine managing 372 nodes, 869 edges and centrality metrics.
- **SQLite WAL + Blockchain Hash Chain** – Thread-safe data storage with SHA-256 evidence sealing.
- **Agency Swarm & spaCy NLP** – CEO Prime Agent multi-agent orchestration & automated IndianLawNER entity extraction.

Statutory Legal Vault :

- **ReportLab Forensic Engine** – Generates court-certified Section 63 BSA dossiers with QR code verification.

React 19

Vite & Tailwind UI

TypeScript

Type-safe Dashboard

Prime Agent

Agency Swarm Core

FastAPI

Python 3.12 REST API

NetworkX

MultiDiGraph Analytics

Leaflet.js

Geo-Intel & CDR Towers

SQLite WAL

Blockchain Ledger

ReportLab

Sec 63 BSA PDF Dossier

Technical Feasibility

- Uses proven open-source frameworks – FastAPI, React 19, NetworkX
- Direct compatibility with police CCTNS and telecom CDR exports
- Prime Agent multi-agent swarm runs efficiently on standard workstation CPUs / GPUs
- Cryptographic SHA-256 ledger ensures 100% zero-gas data integrity

Economic Feasibility

- Zero expensive software licenses or cloud vendor lock-in
- Directly leverages existing police CCTNS infrastructure, telecom CDR logs & standard department PCs
- Compresses weeks of manual investigative analysis into seconds
- Drastically reduces police manpower costs during major syndicate probes

“The solution is technically robust, low-cost, and operationally practical—enabling law enforcement to dismantle organized crime syndicates faster through autonomous multi-agent intelligence.”

Challenges & Strategies

- **Unstructured Police Data** – FIRs contain spelling mistakes, slang, and varied dialects. **Solved via AI Named Entity Recognition + Jaro-Winkler fuzzy deduplication (>88%).**
- **Massive CDR & Financial Scale** – Correlating millions of call records without lag. **Solved via in-memory graph caching & SQLite WAL indexing for sub-millisecond lookups.**

Law Enforcement & Police Officers

- 1-click network expansion reveals 2-hop associates, burner phones, and money trails.
- Automatically identifies shadow kingpins who never make direct phone calls.
- Collaborative case board eliminates inter-departmental communication delays.

Intelligence Analysts & Investigators

- CEO Prime Agent automates tedious cross-referencing across FIRs, CDRs, and bank ledgers.
- Multi-agent swarm auto-calculates threat scores (0-100) and uncovers hidden Hawala laundering loops.
- Cross-case linking surfaces shared burner phones, vehicles, and safehouses across jurisdictions.

Judiciary & Legal Prosecution

- 100% tamper-evident digital evidence meeting Section 63 BSA / 65B IEA standards.
- Court-ready forensic PDF dossiers accelerate trials and eliminate chargesheet delays.

“SENTINEL transforms fragmented crime records into real-time operational clarity—unmasking shadow kingpins, disrupting criminal syndicates, and accelerating the delivery of justice.”

THANK YOU

SENTINEL v2.0 – AI-Powered Criminal Network Analysis System

Team Stryker • Problem Statement ID: 26189

Open for Questions & Live System Demonstration